6

Independence, Basis and Dimension

Historical Note

That a plane is two-dimensional and space three-dimensional is older than mathematics. That the number is *well defined*, that no clever choice of vectors could ever coordinatise the plane with three genuinely independent directions or with one, is a theorem, and it was surprisingly slow to be isolated. Grassmann had the essential exchange argument in the 1862 edition of the *Ausdehnungslehre*; the clean statement we use below is usually credited to Ernst Steinitz, who needed it in his 1910 memoir on the algebraic theory of fields, where the vectors were elements of one field regarded over a smaller one. Five years before that, Georg Hamel had proved something stranger still: assuming the axiom of choice, $\mathbb{R}$ regarded as a vector space over $\mathbb{Q}$ has a basis. Nobody has ever written one down, and nobody can, but its existence is enough to produce a function satisfying $f(x+y) = f(x) + f(y)$ whose graph is dense in the plane. Bases, it turns out, are cheap to prove and expensive to exhibit.

Chapter 5 produced three subspaces from every matrix and described each of them by a handful of generators. It could not say whether the handful was the right size. This chapter supplies the missing notion: a set of generators is *right* when none of them is redundant, such a set is called a basis, and the theorem that makes the whole subject work is that every basis of a given space has the same number of elements.

6.1 Linear independence

Definition 6.1 (Linear independence)**.** *Vectors $v_1, \dots, v_k$ in a vector space V are* linearly independent *if the only scalars satisfying*

$$\lambda_1 v_1 + \lambda_2 v_2 + \cdots + \lambda_k v_k = \mathbf{0}$$

are $\lambda_1 = \cdots = \lambda_k = 0$. Otherwise they are dependent*, and any choice of scalars, not*

all zero, for which the sum vanishes is called a relation *among them. An infinite set is independent if every finite subset of it is.*

Proposition 6.2 (Three readings of independence)**.** *For $v_1, \dots, v_k \in V$ the following are equivalent.*

(a) The v_i are linearly independent.

(b) No v_i lies in the span of the others.

(c) Every vector of span $\{v_1, \dots, v_k\}$ *is a linear combination of the v_i in exactly one way.*

Proof. (a) $\Rightarrow$ (b). If $v_i = \sum_{j \neq i} \mu_j v_j$ then moving everything to one side gives $\sum_{j \neq i} \mu_j v_j + (-1)v_i = \mathbf{0}$, a relation whose ith coefficient is $-1 \neq 0$.

(b) $\Rightarrow$ (a). Suppose $\sum_j \lambda_j v_j = \mathbf{0}$ with some $\lambda_i \neq 0$. Dividing by λ_i, which is legitimate because the scalars form a field, gives $v_i = -\sum_{j \neq i} (\lambda_j / \lambda_i) v_j$, so v_i is in the span of the others.

(a) $\Rightarrow$ (c). If $\sum_j \lambda_j v_j = \sum_j \mu_j v_j$ then $\sum_j (\lambda_j - \mu_j) v_j = \mathbf{0}$, so every $\lambda_j - \mu_j$ vanishes.

(c) $\Rightarrow$ (a). The vector $\mathbf{0}$ is the combination with all coefficients zero; if uniqueness holds, no other combination can equal it, which is exactly (a). $\square$

Key Idea

Independence is a statement about *redundancy*, not about direction. A dependent set carries a vector you could delete without shrinking the span, and reading (b) is the one to keep in mind when you want intuition. Reading (c) is the one that matters in practice, because it is what makes coordinates well defined in Section 6.6.

Caution

Two easy traps. First, any set containing $\mathbf{0}$ is dependent, since $1 \cdot \mathbf{0} = \mathbf{0}$ is a relation with a nonzero coefficient. Second, reading (b) says no vector lies in the span of the *others*, which is weaker than saying no two of them are parallel: the three vectors in Example 6.3(iii) are pairwise non-parallel and still dependent. Checking pairs is never enough beyond $k = 2$.

Example 6.3 (Independent and dependent sets). *(i)* $(1,2)$ *and* $(3,1)$ *in* $\mathbb{R}^2$ *are independent:* $\lambda(1,2)+\mu(3,1)=(0,0)$ *forces* $2\lambda+\mu=0$ *and* $\lambda+3\mu=0$*, hence* $\mu=-2\lambda$ *and* $\lambda-6\lambda=0$*, so* $\lambda=\mu=0$*.*

(ii) $(1,2)$ *and* $(2,4)$ *are dependent, with relation* $2(1,2)-1(2,4)=\mathbf{0}$*. This is Example 2.7 again.*

(iii) $(1,1,0)$, $(0,1,1)$ *and* $(1,0,-1)$ *in* $\mathbb{R}^3$ *are dependent, since*

$$-1\,(1,1,0)+1\,(0,1,1)+1\,(1,0,-1)=(0,0,0),$$

yet no two of them are parallel.

(iv) $1, t, t^2$ *in* P_2 *are independent, because a polynomial is the zero function exactly when all its coefficients vanish.*

(v) $\cos$ *and* $\sin$ *are independent in the space of functions on* $\mathbb{R}$*: if* $a\cos t+b\sin t=0$ *for every* t*, then* $t=0$ *gives* $a=0$ *and* $t=\pi/2$ *gives* $b=0$*.*

(vi) 1, $\cos^2$ *and* $\sin^2$ *are dependent, by the Pythagorean identity* $\cos^2 t+\sin^2 t=1$*. Dependence among functions can encode a genuine theorem.*

Independence is a null space question

In $\mathbb{R}^n$ the test is mechanical, and we already have the machine.

Proposition 6.4 (Testing independence). *Let* $v_1,\dots,v_k\in\mathbb{R}^n$ *and let* A *be the* $n\times k$ *matrix with these vectors as its columns. Then the* v_i *are independent if and only if* $N(A)=\{\mathbf{0}\}$*, which happens exactly when* A *has* k *pivots. In particular, more than* n *vectors in* $\mathbb{R}^n$ *are always dependent.*

Proof. By Proposition 4.4, $Ax=x_1v_1+\cdots+x_kv_k$, so a relation among the v_i is precisely a nonzero element of $N(A)$. Theorem 3.10(c) says $N(A)=\{\mathbf{0}\}$ exactly when there are no free columns, that is, when all k columns are pivot columns. The last statement is Proposition 3.13: with $k>n$ there are at most n pivots, hence a free column and a nonzero relation. □

Example 6.5 (The test in practice). *Are* $(1,2,3)$, $(0,1,1)$ *and* $(1,4,5)$ *independent? Put them in the columns and eliminate:*

$$\begin{pmatrix}1&0&1\\2&1&4\\3&1&5\end{pmatrix}\longrightarrow\begin{pmatrix}1&0&1\\0&1&2\\0&1&2\end{pmatrix}\longrightarrow\begin{pmatrix}1&0&1\\0&1&2\\0&0&0\end{pmatrix},$$

using $R_2 \to R_2 - 2R_1$, $R_3 \to R_3 - 3R_1$, then $R_3 \to R_3 - R_2$. Two pivots, three columns, so there is a free column and the vectors are dependent. Reading off the relation: setting $x_3 = 1$ gives $x_2 = -2$ and $x_1 = -1$, so $-1\,(1,2,3) - 2\,(0,1,1) + 1\,(1,4,5) = (0,0,0)$, which the reader should check directly. The first two columns, whose positions carry pivots, are independent.

Figure 6.1 says the same thing in the plane, where it can be seen rather than computed: three vectors must be dependent, and deleting the redundant one costs nothing, because the other two rebuild it.

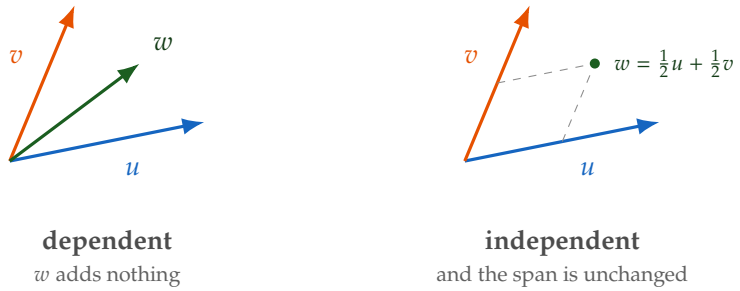


Figure 6.1. Dependence is redundancy. On the left, three vectors in the plane, necessarily dependent by Proposition 6.4 since three exceeds two. On the right, w has been deleted and nothing has been lost: the dashed half-parallelogram rebuilds it from u and v, so the span of the two is the span of the three. A basis is what remains when this deletion can no longer be performed.

6.2 Bases

Definition 6.6 (Basis). *A* basis *of a vector space V is a set of vectors that is linearly independent and spans V. By convention the empty set is a basis of $\{\mathbf{0}\}$.*

Proposition 6.7 (A basis is a unique-representation system). *The vectors $v_1, \dots, v_n$ form a basis of V if and only if every $v \in V$ can be written as $v = \lambda_1 v_1 + \cdots + \lambda_n v_n$ in exactly one way.*

Proof. Spanning says every v has at least one such expression; independence says, by Proposition 6.2(c), that it has at most one. Conversely, existence of an expression for every v is spanning, and uniqueness gives independence by the same proposition. □

Example 6.8 (Bases we already know). *(i) The* standard basis $e_1, \dots, e_n$ *of* $\mathbb{R}^n$, *where* e_i *has a* 1 *in position* i *and zeros elsewhere. Independence and spanning are both immediate, since* $x = x_1e_1 + \cdots + x_ne_n$ *and this is the only such expression.*

(ii) $1, t, t^2, \dots, t^n$ *in* P_n, *by Example 6.3(iv).*

(iii) The matrix units E_{ij} *in* $M_{m\times n}(\mathbb{R})$, *each with a single* 1 *in position* (i, j). *There are* mn *of them.*

(iv) cos *and* sin *in the solution space of* $y'' + y = 0$. *Independence is Example 6.3(v); that they span is the existence and uniqueness theorem for that differential equation, which we take on trust here.*

(v) 1 *and* i *in* $\mathbb{C}$ *regarded as a real vector space.*

(vi) $(1, 1)$ *and* $(1, -1)$ *in* $\mathbb{R}^2$. *There is nothing canonical about the standard basis, and Section 6.6 is about what changes when we choose another.*

6.3 The exchange lemma

Everything now rests on one comparison: an independent set can never be larger than a spanning set. The proof is a replacement procedure, feeding the independent vectors into the spanning set one at a time and throwing out a spanning vector at each step.

Lemma 6.9 (Steinitz exchange). *Suppose* $v_1, \dots, v_m$ *are linearly independent in* V *and* $w_1, \dots, w_n$ *span* V. *Then* $m \le n$, *and after reordering the* w_j *the set* $\{v_1, \dots, v_m, w_{m+1}, \dots, w_n\}$ *still spans* V.

Proof. We prove by induction on r that for each r with $0 \le r \le m$ we have $r \le n$ and, after reordering the w_j,

$$\operatorname{span}\{v_1, \dots, v_r, w_{r+1}, \dots, w_n\} = V.$$

For $r = 0$ this is the hypothesis that the w_j span.

Assume it for some $r < m$. Since the displayed set spans V, we may write

$$v_{r+1} = \sum_{i\le r} a_i v_i + \sum_{j>r} b_j w_j$$

for some scalars. Not every b_j can be zero: if they all were, then v_{r+1} would be a combination of $v_1, \dots, v_r$ alone, and that contradicts the independence of the v_i, by reading (b) of Proposition 6.2. So at least one w_j with $j > r$ is still present, whence $r < n$, that is $r + 1 \le n$. Reorder so that $b_{r+1} \neq 0$, and solve:

$$w_{r+1} = \frac{1}{b_{r+1}}\Big(v_{r+1} - \sum_{i\le r} a_i v_i - \sum_{j>r+1} b_j w_j\Big).$$

So w_{r+1} lies in the span of $v_1, \dots, v_{r+1}, w_{r+2}, \dots, w_n$. That set therefore spans everything the previous one spanned, namely V, which is the statement for $r+1$.

Taking $r = m$ gives $m \le n$ and the final assertion. $\square$

Theorem 6.10 (Dimension is well defined). *If V has a finite basis, then every basis of V is finite and any two bases have the same number of elements.*

Proof. Let B be a finite basis with n elements and let B' be any other basis. Every finite subset of B' is independent and B spans, so by Lemma 6.9 every finite subset of B' has at most n elements; hence B' is finite, say with m elements. Now apply the lemma twice: B' independent against B spanning gives $m \le n$, and B independent against B' spanning gives $n \le m$. $\square$

Definition 6.11 (Dimension). *If V has a finite basis, V is called* finite-dimensional *and its* dimension $\dim V$ *is the number of elements of any basis. Otherwise V is* infinite-dimensional. *We set* $\dim\{\mathbf{0}\} = 0$.

Example 6.12 (Dimensions). *From Example 6.8:* $\dim \mathbb{R}^n = n$; $\dim P_n = n+1$, *one more than the degree because of the constant term;* $\dim M_{m\times n}(\mathbb{R}) = mn$; *the solution space of $y'' + y = 0$ has dimension 2. The spaces $\mathbb{R}[t]$ of all polynomials and $C[0,1]$ of continuous functions are infinite-dimensional, since $1, t, \dots, t^N$ are independent for every N.*

Caution

Dimension depends on the field of scalars, and the notation hides it. The complex numbers form a vector space of dimension 2 over $\mathbb{R}$, with basis $1, i$, and of dimension 1 over $\mathbb{C}$, with basis 1. Write $\dim_{\mathbb{R}} \mathbb{C} = 2$ and $\dim_{\mathbb{C}} \mathbb{C} = 1$ when the distinction matters. In the same spirit $\mathbb{R}$ over $\mathbb{Q}$ is infinite-dimensional, which is the setting of Hamel's theorem in the opening note.

6.4 What dimension buys

Lemma 6.13 (Adding an outside vector preserves independence). *If $v_1, \dots, v_k$ are independent and $w \notin \operatorname{span}\{v_1, \dots, v_k\}$, then $v_1, \dots, v_k, w$ are independent.*

Proof. Suppose $\sum_i \lambda_i v_i + \mu w = \mathbf{0}$. If $\mu \neq 0$ then $w = -\mu^{-1} \sum_i \lambda_i v_i$ lies in the span of the

v_i, contrary to hypothesis. So $\mu = 0$, and then all $\lambda_i = 0$ by independence of the v_i. □

Figure 6.2 follows one list as it grows. Each new vector either raises the dimension of the span by exactly one or leaves it alone, and the second is what happens the moment the new vector was already reachable, which is the moment independence is lost.

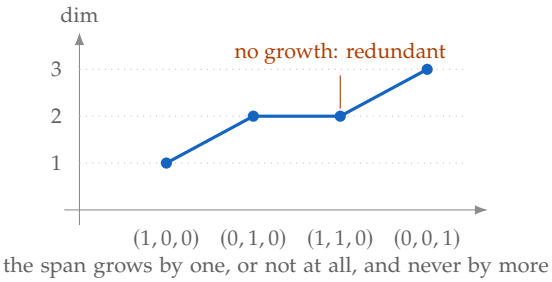


Figure 6.2. Adding a vector to a list either raises the dimension of its span by exactly one or leaves it alone, and the second happens precisely when the new vector was already reachable. Here $(1, 1, 0)$ is the sum of the two before it, so the span does not grow, and the list stops being independent at that moment. That single rule is what makes dimension well defined: any two ways of building up to the same span must take the same number of productive steps, which is Theorem 6.14. It is also the algorithm. Feeding vectors in one at a time and keeping only those that raise the dimension is how a basis is extracted from a spanning set in practice, and Gaussian elimination is that procedure with the bookkeeping written as a matrix.

Theorem 6.14 (Counting in a finite-dimensional space)**.** *Let* $\dim V = n$.

(a) Any $n + 1$ or more vectors in V are dependent.
(b) No set of fewer than n vectors spans V.
(c) Any n independent vectors form a basis.
(d) Any n vectors that span V form a basis.
(e) Every independent set in V can be extended to a basis.
(f) Every finite spanning set of V contains a basis.
(g) If $W \subseteq V$ is a subspace then $\dim W \leq n$, with equality only for $W = V$.

Proof. Fix a basis B with n elements.

(a) and (b) are Lemma 6.9, applied with B as the spanning set and as the independent set respectively.

(c) Let $v_1, \dots, v_n$ be independent. If they did not span, pick $w \notin \operatorname{span}\{v_1, \dots, v_n\}$; by Lemma 6.13 we would have $n + 1$ independent vectors, contradicting (a).

(d) Let $v_1, \dots, v_n$ span. If they were dependent, some v_i would lie in the span of the others by Proposition 6.2(b), and deleting it would leave $n-1$ vectors still spanning V, contradicting (b).

(e) Given an independent set S, repeatedly apply Lemma 6.13: while S does not span, adjoin a vector outside its span. Each step keeps S independent and increases its size, and by (a) the process must stop within n steps, at which point S spans and is a basis.

(f) Given a finite spanning set, repeatedly delete any vector lying in the span of the others. Each deletion leaves the span unchanged, and the process stops because the set is finite; what remains spans and is independent by Proposition 6.2(b).

(g) Any independent subset of W is independent in V, so by (a) it has at most n elements. Choose one, say S, of the largest possible size $k \le n$. If S did not span W, there would be some $w \in W$ outside its span, and $S \cup \{w\}$ would be independent by Lemma 6.13, contradicting maximality. So S is a basis of W and $\dim W = k \le n$. If $k = n$ then S is a set of n independent vectors in V, hence a basis of V by (c), so $V = \operatorname{span} S \subseteq W$. □

Figure 6.3 is the whole theorem in one picture: independence bounds a set from above, spanning bounds it from below, and the two bounds meet at a single number.

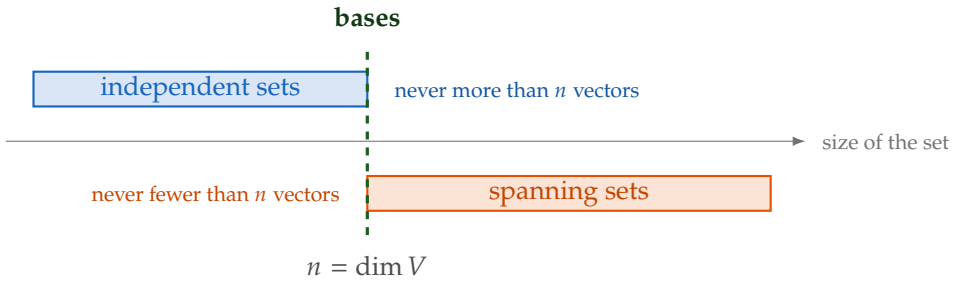


Figure 6.3. Theorem 6.14 as a picture. Independence bounds a set from above and spanning bounds it from below, and the two bounds meet at a single number. A basis is precisely a set that is simultaneously as large as an independent set can be and as small as a spanning set can be, which is why parts (c) and (d) let you verify only half the definition once you know the dimension.

Key Idea

Parts (c) and (d) are the ones you will use daily. Once the dimension is known, checking that n vectors form a basis requires verifying independence *or* spanning, never both. In practice independence is the cheaper test, because Proposition 6.4 turns it into one elimination.

6.5 Bases for the subspaces of a matrix

We can now say exactly how large the three subspaces of Chapter 5 are, and the answer is the same number three times over.

Theorem 6.15 (Bases from one elimination). *Let A be $m \times n$ with r pivots, and let U be any row echelon form of A.*

(a) The r nonzero rows of U form a basis of $R(A)$, so $\dim R(A) = r$.

(b) The r columns of the original *A in the pivot positions form a basis of $C(A)$, so $\dim C(A) = r$.*

(c) The $n - r$ special solutions, one obtained by setting each free unknown to 1 and the others to 0, form a basis of $N(A)$, so $\dim N(A) = n - r$.

Proof. (a) The nonzero rows span $R(A)$ by Proposition 5.13. They are independent: suppose a combination of them vanishes, and let the pivots sit in columns $j_1 < \cdots < j_r$. Looking at column j_1, every row below the first has a zero there, so the coefficient of the first row must be zero. Looking then at column j_2, the same argument kills the coefficient of the second row, and so on down.

(b) By Lemma 4.20 the pivot columns of the reduced form are exactly those not expressible through their predecessors, and by Lemma 4.19 the columns of A satisfy precisely the same relations. So the pivot columns of A are independent, and every other column of A is a combination of the pivot columns to its left; hence they span $C(A)$.

(c) Let F be the set of indices of the free columns, so $|F| = n - r$, and for each $j \in F$ let $s_j \in N(A)$ be the solution obtained by setting $x_j = 1$, setting the other free unknowns to 0, and back substituting. The construction in the proof of Theorem 3.10 shows that every element of $N(A)$ is $\sum_{j \in F} x_j s_j$, where the x_j are its own free entries, so the s_j span. They are independent because s_j has a 1 in position j and every other $s_{j'}$ has a 0 there: a vanishing combination read at position j gives the coefficient of s_j as 0. □

Definition 6.16 (Rank and nullity). *The* rank *of A is* $\operatorname{rank} A = \dim C(A)$, *and the* nullity *is* $\dim N(A)$.

Corollary 6.17 (Rank counted three ways). *For any $m \times n$ matrix A,*

$$\operatorname{rank} A = \dim C(A) = \dim R(A) = \textit{the number of pivots}, \qquad \operatorname{rank} A + \dim N(A) = n.$$

In particular the row rank and the column rank of a matrix are equal, and $\operatorname{rank} A = \operatorname{rank} A^t$.

Proof. The three descriptions of r are Theorem 6.15(a),(b) together with Theorem 3.9, which says the number of pivots depends only on A. Adding (b) and (c) gives $r + (n - r) = n$. Finally $R(A)$ is spanned by the rows of A, which are the columns of A^t, so $\dim R(A) = \operatorname{rank} A^t$. □

Insight

That row rank equals column rank is the first genuinely surprising theorem in the subject. The rows of A live in $\mathbb{R}^n$ and the columns in $\mathbb{R}^m$; the two spans sit in different spaces, can have nothing to do with each other geometrically, and are described by completely different data. Yet they always have the same dimension, because both are counting the same pivots. When we reach the singular value decomposition, this equality becomes the statement that a matrix has one list of singular values whichever way you look at it, and it is what makes low-rank approximation meaningful: rank is a property of the matrix, not of the direction from which you read it.

Example 6.18 (Dimensions, from the elimination we already did)**.** *Return to Example 5.14, with*

$$A = \begin{pmatrix} 1 & 2 & 0 & 1 \\ 2 & 4 & 1 & 4 \\ 3 & 6 & 1 & 5 \end{pmatrix}, \qquad m = 3, \quad n = 4.$$

The elimination gave two pivots, in columns 1 *and* 3*. So* $\operatorname{rank} A = 2$*, and* $\dim R(A) = \dim C(A) = 2$ *while* $\dim N(A) = 4 - 2 = 2$*, which matches* $\operatorname{rank} A + \dim N(A) = 4$*. The generators found there were bases all along, and we now know they were not merely convenient but minimal: no single vector could span* $C(A)$*, because the two columns* $(1, 2, 3)$ *and* $(0, 1, 1)$ *are not parallel.*
Note also what $\operatorname{rank} A = 2 < 3 = m$ *says:* $C(A)$ *is a plane inside* $\mathbb{R}^3$*, not all of it, so* $Ax = b$ *is inconsistent for most* b*. And* $\dim N(A) = 2 > 0$ *says that when it is consistent, the solution set is a two-parameter family.*

6.6 Coordinates

A basis does more than count. It turns an abstract space into a concrete one.

Definition 6.19 (Coordinate vector). *Let $B = (v_1, \dots, v_n)$ be an* ordered *basis of V. For $v \in V$, write $v = \lambda_1 v_1 + \cdots + \lambda_n v_n$, which is possible in exactly one way by Proposition 6.7, and define the* coordinate vector

$$[v]_B = (\lambda_1, \dots, \lambda_n) \in \mathbb{R}^n.$$

Proposition 6.20. *The map $v \mapsto [v]_B$ is a bijection $V \to \mathbb{R}^n$ satisfying $[u+v]_B = [u]_B + [v]_B$ and $[\lambda v]_B = \lambda [v]_B$.*

Proof. It is well defined and injective by uniqueness of the representation, and surjective because any $(\lambda_1, \dots, \lambda_n)$ is the coordinate vector of $\sum_i \lambda_i v_i$. If $u = \sum \alpha_i v_i$ and $v = \sum \beta_i v_i$ then $u + v = \sum (\alpha_i + \beta_i) v_i$ and $\lambda v = \sum (\lambda \beta_i) v_i$, and by uniqueness those are *the* representations. □

Key Idea

Every n-dimensional real vector space is $\mathbb{R}^n$ in disguise, and choosing a basis is precisely the act of removing the disguise. This is why it is legitimate to prove theorems about $\mathbb{R}^n$ and then use them about polynomials, matrices or solutions of a differential equation.

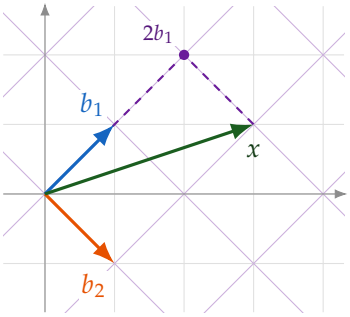
Caution

The disguise is removed by a choice, and different choices give different answers. The vector does not change; its coordinates do. Almost every confusion in the next few chapters comes from forgetting which basis a list of numbers is written in, which is why the notation carries the subscript B. From here on, whenever a matrix appears, it is worth asking which basis it is expressed in.

Example 6.21 (Reading the same vector twice). *Let $B = \big((1,1),(1,-1)\big)$, which is a basis of $\mathbb{R}^2$ by Theorem 6.14(c), since the two vectors are independent and $\dim \mathbb{R}^2 = 2$. To find $[x]_B$ for $x = (3,1)$, solve*

$$a(1,1) + b(1,-1) = (3,1) \iff a + b = 3, \quad a - b = 1 \implies a = 2,\ b = 1.$$

So $[x]_B = (2,1)$ while $[x]_E = (3,1)$ in the standard basis E. Two different pairs of numbers, one vector.



$x = 3e_1 + 1e_2$
$x = 2b_1 + 1b_2$
so $[x]_E = (3, 1)$
and $[x]_B = (2, 1)$

Figure 6.4. One vector, two coordinate systems. The grey squares are the standard grid and the purple diamonds are the grid of $B = \big((1, 1), (1, -1)\big)$. The arrow x is drawn once and never moves; only the list of numbers describing it changes, from $(3, 1)$ in the standard basis to $(2, 1)$ in B. The dashed route is the B-reading: two steps along b_1, then one along b_2.

In P_2 with the basis $B' = (1, t, t^2)$, the polynomial $p(t) = 5 - 3t + t^2$ has $[p]_{B'} = (5, -3, 1)$. Change the basis to $B'' = \big(1, t - 1, (t - 1)^2\big)$, which is also a basis. Substituting $t = s + 1$,

$$p(s + 1) = 5 - 3(s + 1) + (s + 1)^2 = s^2 - s + 3, \qquad \text{so} \qquad p(t) = 3 - (t - 1) + (t - 1)^2$$

and $[p]_{B''} = (3, -1, 1)$. The polynomial has not changed; the coordinates have, and the second reading is the one a numerical analyst would want if the interesting behaviour happens near $t = 1$.

Open Problem 6.22 (Rota's basis conjecture)**.** *Let V be an n-dimensional vector space and let $B_1, \ldots, B_n$ be n bases of it, written as the rows of an $n \times n$ array so that row i holds the n vectors of B_i.* Can the array always be rearranged within its rows so that every column is also a basis?

What is known. *Nothing resembling a general proof, despite the statement being about the most basic object in the chapter. It is known for $n \leq 4$ by exhaustive argument, for vector spaces over fields of characteristic zero when n is one more or one less than a prime, and asymptotically: one can always find $n - O(\sqrt{n})$ disjoint transversal bases rather than all n. The conjecture is due to Rota, and was circulated in the form above by Huang and Rota [340]. It is one of a small family of statements, including the Alon and Tarsi conjecture on Latin squares, that are known to be equivalent or nearly so, and none of them has yielded. Those two facts are connected, and the connection explains the odd looking list of dimensions: Huang and Rota showed that the Alon and Tarsi conjecture for n implies this one in characteristic zero, that conjecture is a statement about* even *n, and*

it was settled by Drisko at $n = p + 1$ and by Glynn at $n = p - 1$ for odd primes p. Both of those are even, so every dimension the argument reaches is even, and no odd dimension above 3 is known at all.

Where the difficulty sits. *Any single column is easy: a greedy exchange argument, of the kind used to prove the exchange lemma in this chapter, produces one transversal basis with no trouble. The difficulty is that the columns compete. Every greedy proof produces a first column that makes some later column impossible, and no ordering principle is known that avoids this.* The conjecture is a statement that n local exchange arguments can be run simultaneously*, and the exchange lemma gives no purchase on simultaneity.*

Where a reader might start. *Do the case $n = 2$ on paper: two bases of a plane, four vectors, and a rearrangement to find. Then try $n = 3$ with three bases of $\mathbb{R}^3$ chosen adversarially, and notice how quickly the greedy choice paints itself into a corner. That failure is the whole problem. The matroid generalisation, which is the form most current work takes, is in Chapter 66.*

Exercises

Exercise 6.23. *Decide whether each set is independent, and exhibit a relation when it is not.*

(a) $(1,0,1), (0,1,1), (1,1,0)$ *in* $\mathbb{R}^3$.

(b) $(1,2,3), (4,5,6), (7,8,9)$ *in* $\mathbb{R}^3$.

(c) $1 + t,\ 1 - t,\ t^2$ *in* P_2.

(d) $\left(\begin{smallmatrix}1&0\\0&1\end{smallmatrix}\right), \left(\begin{smallmatrix}0&1\\1&0\end{smallmatrix}\right), \left(\begin{smallmatrix}1&1\\1&1\end{smallmatrix}\right)$ *in* $M_2(\mathbb{R})$.

Exercise 6.24. *Find a basis and the dimension of each subspace.*

(a) $\{(x,y,z) \in \mathbb{R}^3 : x + y + z = 0\}$.

(b) The symmetric 2×2 real matrices.

(c) The polynomials $p \in P_3$ with $p(1) = 0$.

(d) $\operatorname{span}\{(1,1,0,0), (0,1,1,0), (1,2,1,0)\}$ *in* $\mathbb{R}^4$.

Exercise 6.25. *For* $A = \left(\begin{smallmatrix}1&2&3&1\\2&4&7&4\\1&2&4&3\end{smallmatrix}\right)$*, find bases for $R(A)$, $C(A)$ and $N(A)$, state* $\operatorname{rank} A$*, and check Corollary 6.17 numerically.*

Exercise 6.26. *Let* $B = \big((2,1),(1,3)\big)$. *Verify that* B *is a basis of* $\mathbb{R}^2$ *and compute* $[x]_B$ *for* $x = (5,5)$ *and for* $x = (1,0)$.

Exercise 6.27. *Extend* $(1,1,0,0)$ *and* $(0,1,1,0)$ *to a basis of* $\mathbb{R}^4$, *following the procedure in the proof of Theorem 6.14(e). How many different answers are possible?*

Exercise 6.28. *Let* $\dim V = 4$ *and let* $S \subseteq V$ *have four elements. Explain why checking that* S *is independent is enough to conclude that* S *is a basis. Then show that the count cannot be dropped: give four independent vectors in* $\mathbb{R}^5$ *that are not a basis, and five vectors that span* $\mathbb{R}^4$ *and are not a basis.*

Exercise 6.29. *Let* A *be* $m \times n$. *Prove that* $\operatorname{rank} A \le \min(m,n)$, *and that* $\operatorname{rank}(AB) \le \min(\operatorname{rank} A, \operatorname{rank} B)$ *whenever the product is defined.* Hint: for the second part, compare column spaces, then use $\operatorname{rank} A = \operatorname{rank} A^{t}$.

Exercise 6.30. *Show that* $1, \cos t, \cos 2t$ *are independent as functions on* $\mathbb{R}$, *and that* $1, \cos^2 t, \cos 2t$ *are not.* Hint: for the first, evaluate at three well-chosen points; for the second, recall a double-angle identity.

Exercise 6.31. *Let* U *and* W *be subspaces of a finite-dimensional* V *with* $\dim U + \dim W > \dim V$. *Show that* $U \cap W \neq \{\mathbf{0}\}$. Hint: take bases of U and of W and count what Theorem 6.14(a) says about their union.

References and further reading

The exchange lemma is Steinitz's, and the account here is the standard replacement argument; Grassmann's earlier version and the general history of how dimension came to be recognised as a theorem rather than an observation are treated in the historical literature on the subject. Hamel's 1905 construction of a basis for $\mathbb{R}$ over $\mathbb{Q}$ is where the finite-dimensional theory of this chapter visibly stops and set theory begins. Among the textbooks, Axler [37] is the sharpest on exactly which results need finite dimension and which do not, and his treatment of Theorem 6.14 is the one we have followed most closely in spirit, though not in wording. Hefferon [307] gives more worked practice at finding bases for the subspaces of a matrix than we have room for, and Austin [35] is

good on the coordinate picture of Figure 6.4. That row rank equals column rank has many proofs; ours goes through the pivot count, which is the cheapest, while Banica [51] obtains it later as a corollary of the singular value decomposition, which is the most illuminating. Both are worth knowing.